

Importing libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Problem statement

- Netflix wants to decide what kind of shows/movies to produce and how to grow in different countries,We will look at the data on Netflix shows and movies to find patterns and trends. This will help us give useful suggestions for better decisions.

Loading data

```
df = pd.read_csv("netflix.csv")
```

Basic info of shape,data type,missing values and renaming column names for better understanding

```
df.shape      # We have 8,807 rows and 12 columns in our Netflix dataset, and it is a long dataset.
df.info()     # Data types and non null values
df.describe() # Summary of the dataset
df.isnull().sum() # Checking for missing values
df.rename(columns={'listed_in' : 'genre'}, inplace=True) # So it's much clearer and more meaningful to rename it to genre.
```

Non Grahical Analysis

```
df["type"].value_counts() # Shows the count of each type of contentavailable on Netflix.
df["type"].unique()      # show the unique content types present in the dataset.
df["director"].value_counts() # Lists the directors and count of titles of each has directed, helping identify the most common directors.
df["country"].value_counts() # Shows how many titles are produced in each country, helping identify top content-producing countries.
df["rating"].value_counts() # Displays the different ratings and how many titles fall under each rating.
df["genre"].value_counts() # Lists the number of titles in each category (e.g., Comedies, Dramas, Documentaries).
df['release_year'].value_counts().sort_index() # Shows how many titles were released each year, sorted by year to help identify content trends over time.
```

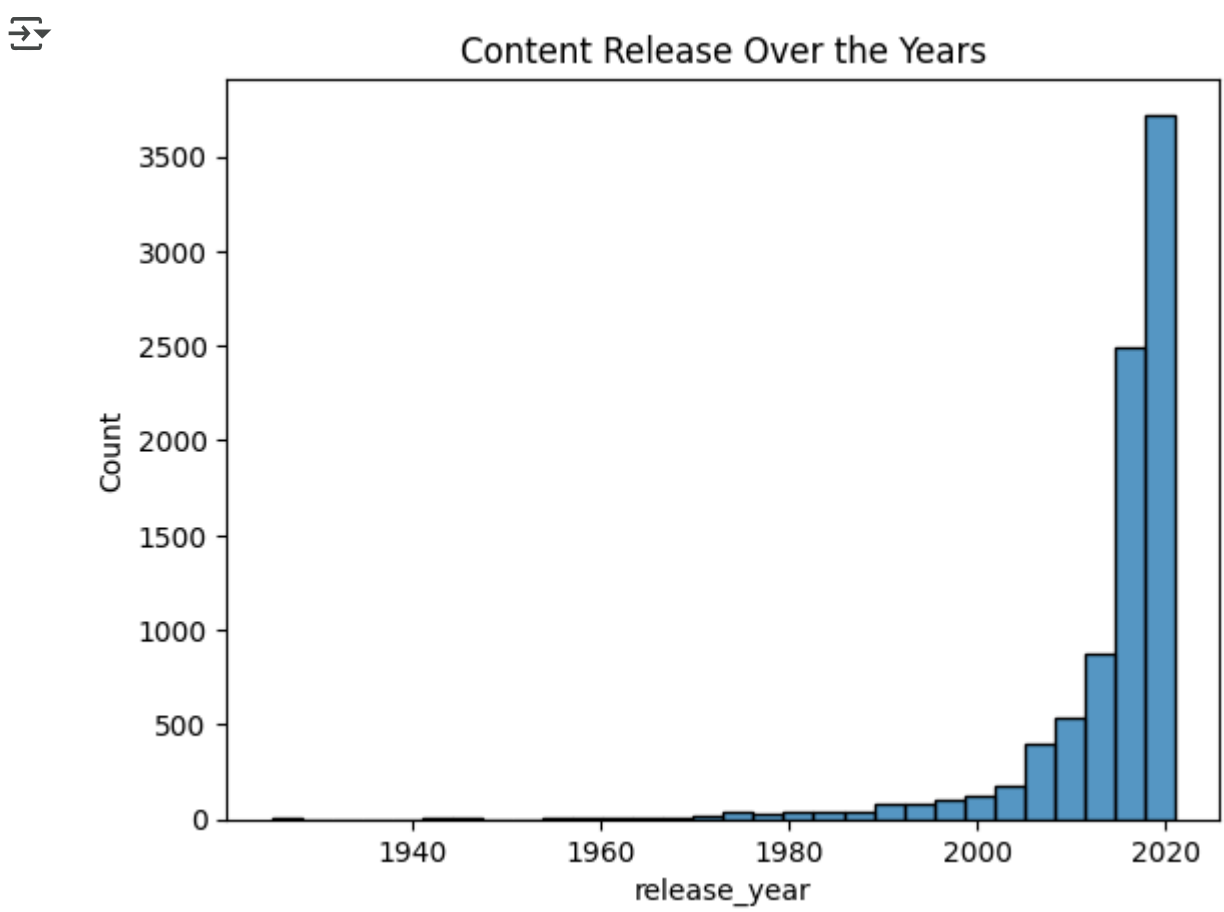
Visual Analysis

4.1 For continuous variable(s): Distplot, countplot, histogram for univariate analysis

4.1(1). Histogram

```
sns.histplot(df['release_year'], bins=30)
plt.title("Content Release Over the Years")
plt.show()

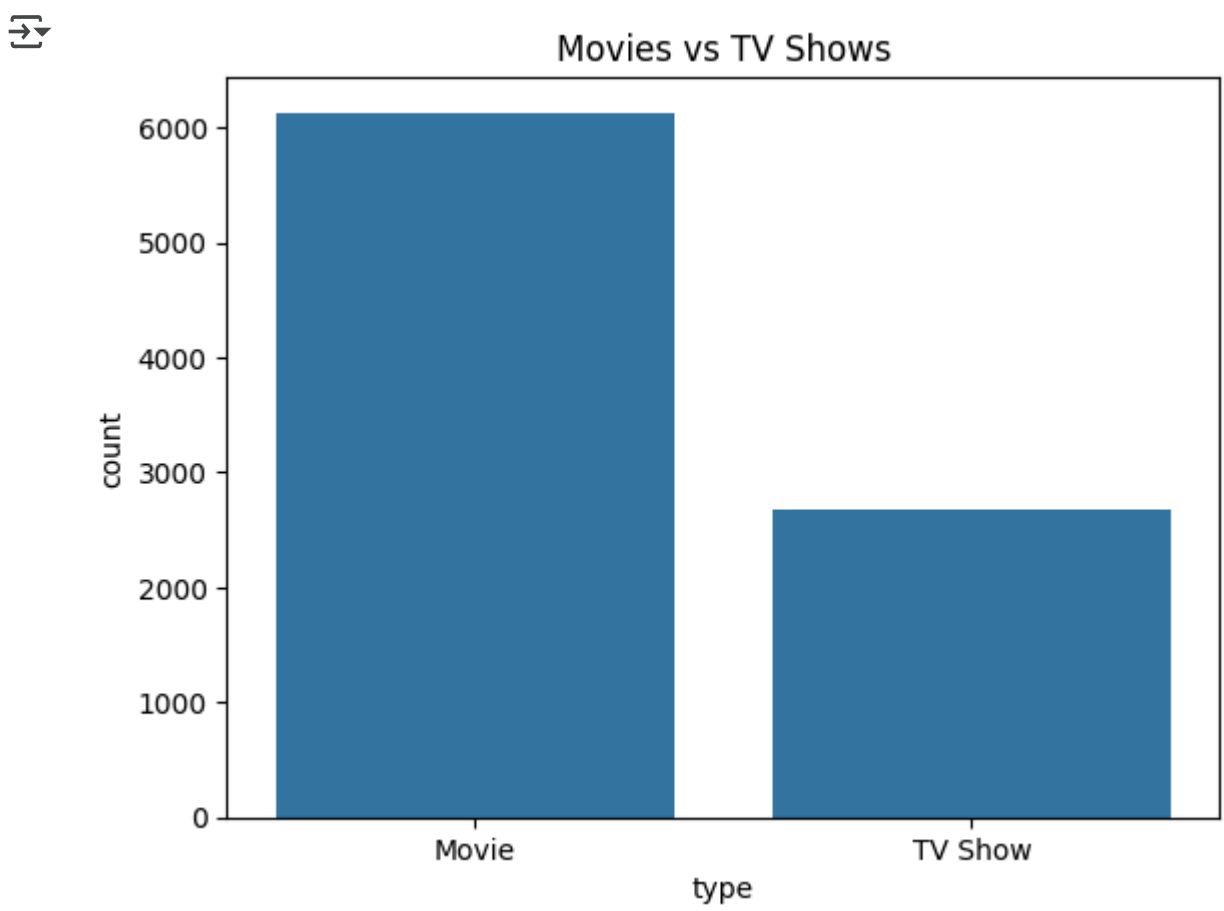
# histogram insights
# - Majority of content is added after 2010, showing rapid expansion.
```



4.1(2).countplot

```
sns.countplot(data=df, x='type')
plt.title("Movies vs TV Shows")
plt.show()

# countplot insight
# - Netflix has more Movies than TV Shows overall.
```



4.2 For categorical variable(s): Boxplot

4.2(1) Boxplot

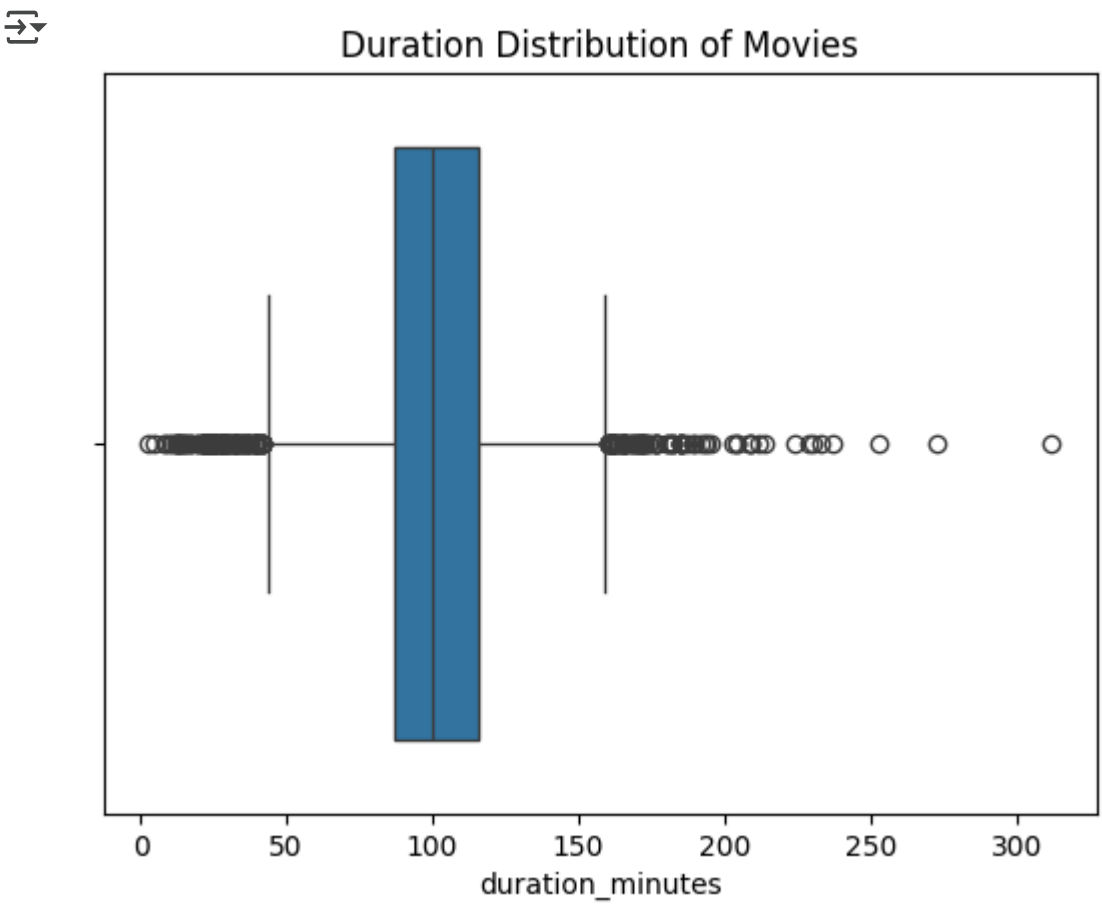
```
movies = df[df['type'] == 'Movie'] # creating new dataframe of movies
movies['duration_minutes'] = movies['duration'].str.extract('(\d+)') # extracting numerical part from duration column as duration column contain "90 min" string like this
movies['duration_minutes'] = pd.to_numeric(movies['duration_minutes'], errors='coerce') # converts the extracted numbers (which are still strings) into numeric type .If a value can't be converted (like if it's blank or contains text), it becomes NaN (missing value), because of
```

```
movies.head(1)
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	genre	description	duration_minutes
0	s1	Movie	Dick Johnson is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020.0	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...	90

```
# Boxplot of duration for Movies
sns.boxplot(data=movies[movies['type'] == 'Movie'], x='duration_minutes')
plt.title("Duration Distribution of Movies")
plt.show()

# Boxplot insight
# - The majority of Netflix movies are between 90 to 110 minutes in duration.
# - The median duration is around 100 minutes.
# - outliers: some movies are shorter than 50 minutes, and a few are extremely long (over 150 minutes and even 300+).
# - outliers indicates that Netflix has a small set of very short and very long movies, but most fall within a standard range.
```

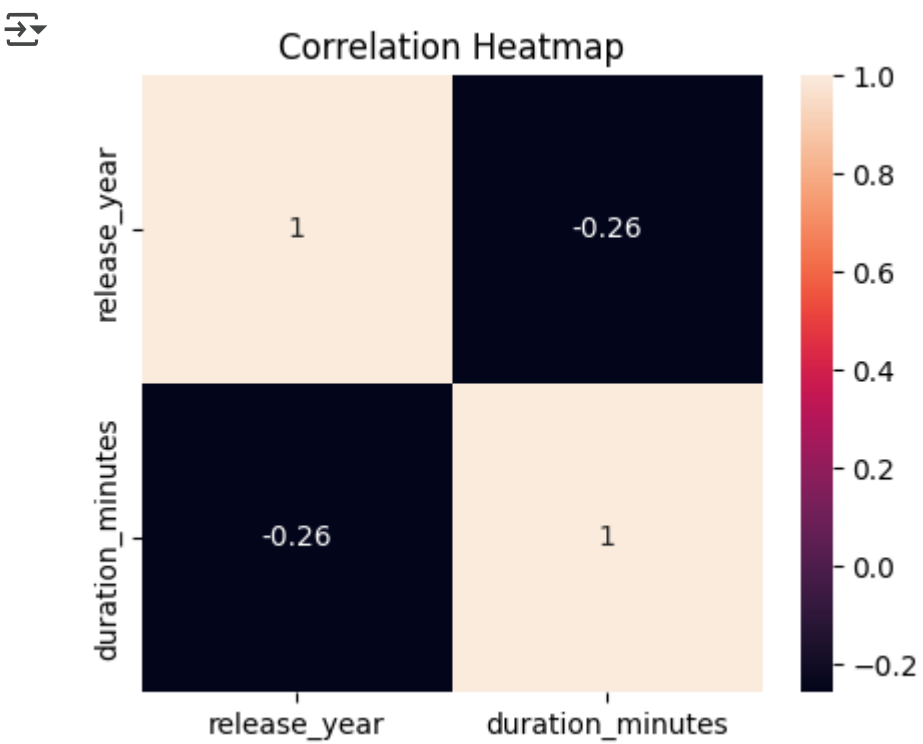



4.3 For correlation: Heatmaps, Pairplots

4.3(1) Heatmap

```
# Heatmap for numeric columns
plt.figure(figsize=(5,4))
sns.heatmap(movies[['release_year', 'duration_minutes']].corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()

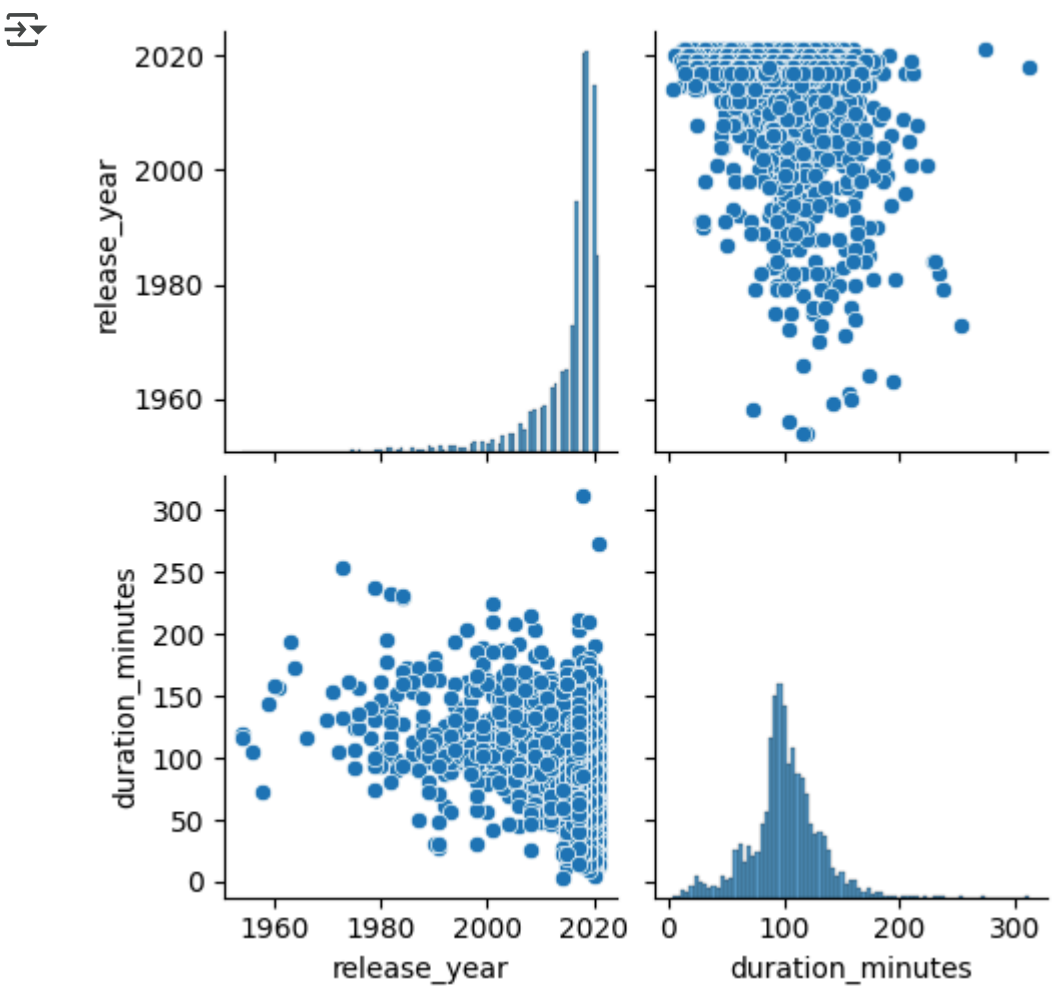
# Heatmap insights
# - The correlation between release year and movie duration is -0.21.
# - This means there is a small negative link – as the release year goes up (i.e., movies are more recent), the length of the movies becomes a bit shorter.
```



4.3(2) pairplot

```
sns.pairplot(movies[['release_year', 'duration_minutes']].dropna())
plt.show()

## pairplot insights
# - Most movies on Netflix were released after the year 2000.
# - Most movies are between 60 to 120 minutes long, which means the usual length is about 1 to 2 hours.
# - A few movies are very long (over 200–300 minutes), which could be special cases or errors in the data.
# - The scatter plot shows movies of all lengths in all years, but in recent years, more movies are in the 80–120 minute range.
```

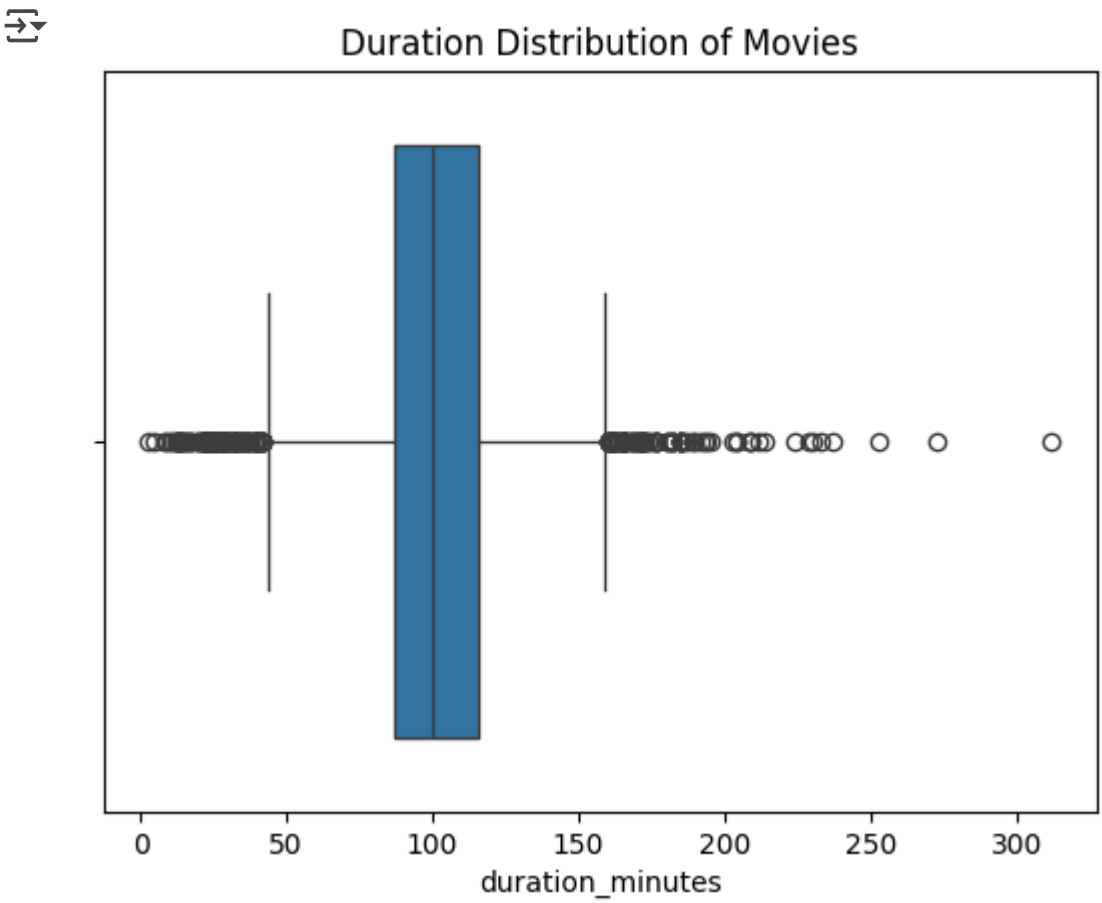


Missing Value & Outlier check

```
#1. Missing values
df.isnull().sum() ## Checks for missing (null) values in each column and returns the total count per column.

#2. Outlier check using boxplot
sns.boxplot(data=movies[movies['type'] == 'Movie'], x='duration_minutes') # It helps to easily detect outliers – values that are much higher or lower than most other data points.
plt.title("Duration Distribution of Movies")
plt.show()

# as we can see some movies are shorter than 50 minutes, and a few are extremely long (over 150 minutes and even 300+).
```

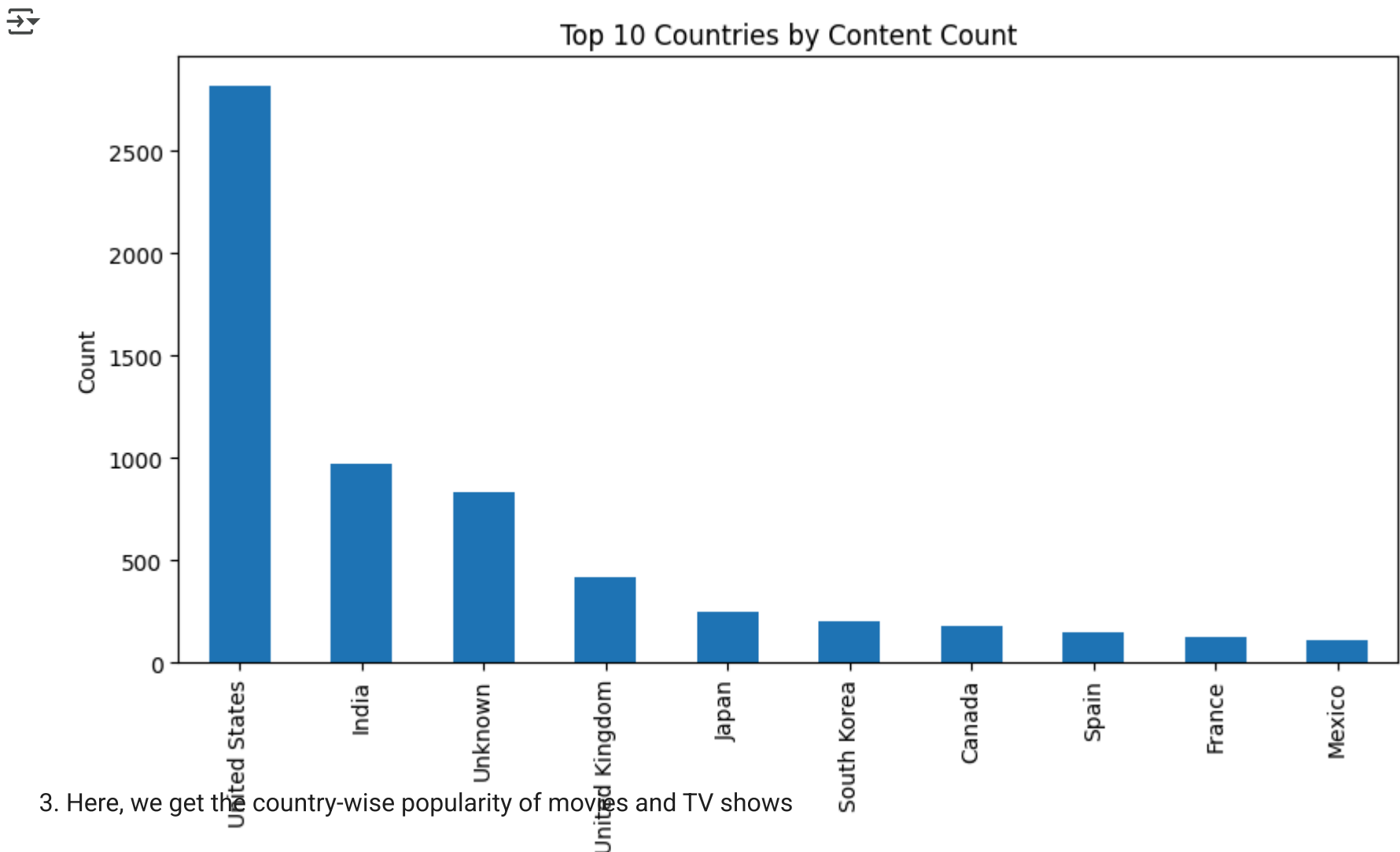


Business Insights

- 1. Histogram show us that Netflix has focused heavily on producing and adding content after 2010, indicating aggressive growth strategy.
- 2. Top 10 countries by content count

```
top_countries = df['country'].value_counts().head(10)

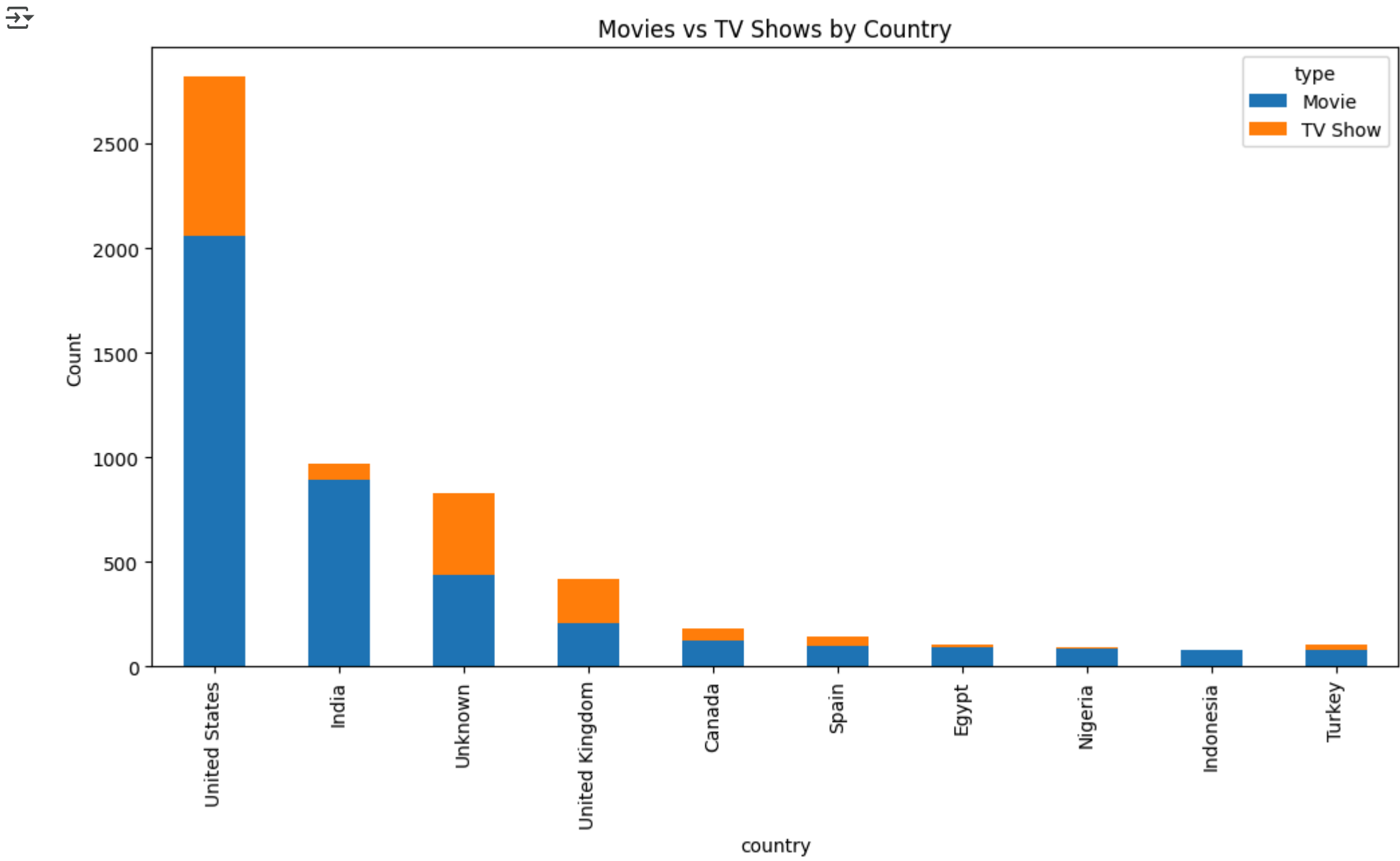
top_countries.plot(kind='bar', figsize=(10, 5), title='Top 10 Countries by Content Count')
plt.xlabel('Country')
plt.ylabel('Count')
plt.xticks(rotation=90)
plt.show()
```

3. Here, we get the country-wise popularity of movies and TV shows

```
df['country'] = df['country'].fillna('Unknown')
country_type = df.groupby(['country', 'type']).size().unstack().fillna(0)
country_type = country_type.sort_values(by='Movie', ascending=False)
country_type.head(10).plot(kind='bar', stacked=True, figsize=(12,6))
plt.title("Movies vs TV Shows by Country")
plt.ylabel("Count")
plt.show()

# Explanation
# - First we fill Nan values with unknown string
# - After that we Groups data by Country and Type (like Movie or TV Show)
# - Size() counts how many items are in each group
# -unstack() turns the "Type" (Movie or TV Show) into columns
# -fillna(0) replaces any missing values with 0
# -Sorts the countries by Movie count in descending order
# -Takes the top 10 countries from the table
# -Creates a stacked bar chart where:
#   One part of the bar shows Movies
#   The other part stacked on top shows TV Shows
```



4. which types of shows or movies netflix should produce to grow a business country wise

```
# Split genre into multiple rows
df_genre = df.assign(genre=df['genre'].str.split(', ').explode('genre'))

# Group by country, type, and genre
grouped = df_genre.groupby(['country', 'type', 'genre']).size().reset_index(name='count')

# Top 3 countries by content count
top_3_countries = df['country'].value_counts().head(1).index.tolist()

# Display top 5 genres for Movies and TV Shows for top 1 countries
for country in top_3_countries:
    print(f"\n=== {country} ===")
    for content_type in ['Movie', 'TV Show']:
        top_genres = grouped[
            (grouped['country'] == country) &
            (grouped['type'] == content_type)
        ].sort_values(by='count', ascending=False).head(5)

        print(f"\nTop 5 genres in {country} for {content_type}:")
        print(top_genres[['genre', 'count']])
```

```
=== United States ===

Top 5 genres in United States for Movie:
   genre  count
2466  Dramas   591
2463  Comedies  524
2465  Documentaries  411
2469  Independent Movies  382
2461  Children & Family Movies  272

Top 5 genres in United States for TV Show:
   genre  count
2493  TV Comedies   234
2494  TV Dramas   189
2483  Docuseries   173
2485  Kids' TV     146
2487  Reality TV   114
```

- Recommendations
1. Focus on producing movies with durations between 90 to 120 minutes to match viewer preferences.
 2. Increase investment in content production for United States and India, where viewership is highest.
 3. Expand movie genres, especially Drama, Comedy, and International Movies, to cater to popular demand.
 4. Differentiate content strategies for Movies and TV Shows based on country-specific trends.
 5. Promote regionally relevant and culturally diverse content to engage local audiences globally.